

ME 315: Manufacturing Engineering-II

Course Code	ME 315
Course Name	Manufacturing Engineering-II
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG
Approval	53rd BoA (Dec 2023)

Course Contents

- **Powder Metallurgy:** Characterization of Engineering Powders, Production of metallic powders, conventional pressing and sintering, alternative pressing and sintering techniques, hot isostatic pressing, metal injection molding, powder injection molding. (7 Lectures)
- **Rapid Prototyping Processes:** Introduction to rapid prototyping and rapid tooling, solid state methods (FDM, LOM), liquid-based (SLA, SGC), powder-based (3DP) RP processes (8 Lectures)
- **Jigs and Fixtures:** Usefulness of Jigs and Fixtures, design principles of jigs and fixtures, principle of location and clamping, types of locating and clamping devices, examples of jigs and fixtures used in lathe machine, milling, boring, shaping, welding and grinding, economics of jigs and fixtures, loading and unloading time, modular fixturing. (7 Lectures)
- **Metrology:** Dimensions and Tolerances, inspection types and principles, radius and taper measurements, measurement of screw threads gears, limits, fits, dimensional tolerances, Conventional measuring instruments and gages, surfaces, measurements of surfaces. (7 Lectures)
- **Integrated Manufacturing Systems:** Material handling, fundamentals of production lines, manual assembly lines, automated production lines, cellular manufacturing, flexible manufacturing systems and cells, computer integrated manufacturing. (7 Lectures).
- **Production Planning and Control:** Aggregate Planning and master production schedule, inventory control, material and capacity requirements planning, Lean production, shop floor control. (6 Lectures)

Text Books

1. Groover, M.P., **Fundamentals of modern manufacturing: materials, processes, and systems**, John Wiley & Sons, 2020.
2. Kalpakjian, S. and Schmid, S.R., **Manufacturing engineering and technology**, Prentice Hall, 2001.

References

None